

Expectation.

Def'n. For a discrete r.v. Y w/ support $S_Y \subseteq \mathbb{R}$ and pmf p_Y ,

we define its

expectation/expected value/mean as

$$\mathbb{E}[Y] := \sum_{y \in S_Y} y \cdot p_Y(y) \quad \text{if the sum exists.}$$

St. Petersburg Paradox

Example. Bernoulli. $Y \sim \mathcal{B}(p)$

$$\mathbb{E}[Y] = ?$$

$$Y \sim \begin{cases} 1 & \text{w/ prob. } p \\ 0 & \text{w/ prob. } q := 1-p \end{cases}$$

$$\mathbb{E}[Y] = 1 \cdot p + 0 \cdot (1-p) = p \quad (\star)$$

Theorem. Let Y_1 and Y_2 be two r.v.s on the same Ω , both w/ finite expectations.

Let α and β be two constants.

Then,

$\mathbb{E}[\alpha \cdot Y_1 + \beta \cdot Y_2]$ also exists, and

$$\mathbb{E}[\alpha Y_1 + \beta Y_2] = \alpha \mathbb{E}[Y_1] + \beta \mathbb{E}[Y_2]$$

Linearity of Expectation.

Example. Binomial. $Y \sim b(n, p)$

$$\mathbb{E}[Y] = ?$$

Start w/ $I_j \sim \mathcal{B}(p)$, $j = 1 \dots n$, **independent**

$$Y = I_1 + I_2 + \dots + I_n$$

$$\mathbb{E}[Y] = \mathbb{E}[I_1 + \dots + I_n] \quad \text{linearity of } \mathbb{E}$$

$$= \mathbb{E}[I_1] + \dots + \mathbb{E}[I_n]$$

$$\stackrel{\star}{=} p + \dots + p = n \cdot p$$

$$\boxed{\mathbb{E}[Y] = n \cdot p}$$

Example. Geometric. $Y \sim g(p)$

$$\mathbb{E}[Y] = ?$$

Differentiate
Power Series.

Idea #1. $\mathbb{E}[Y] = \sum_{k=0}^{\infty} k \cdot \underbrace{p_Y(k)}_{\text{circled}} = \sum_{k=0}^{\infty} k \cdot q^k \cdot p = p q \sum_{k=1}^{\infty} k \cdot q^{k-1}$

Idea #2. $\sum_{k=1}^{\infty} k \cdot p_k = p_1 + 2 \cdot p_2 + 3 p_3 + \dots + k \cdot p_k + \dots$

$$\begin{aligned} &= p_1 + \\ &\quad + p_2 + p_2 + \\ &\quad + p_3 + p_3 + p_3 \\ &\quad \vdots \\ &\quad + p_k + p_k + p_k + \dots + p_k \\ &\quad \vdots \\ &= \mathbb{P}[Y > 0] + \mathbb{P}[Y > 1] + \dots + \mathbb{P}[Y > k-1] + \dots \end{aligned}$$

$$\mathbb{E}[Y] = \sum_{k=0}^{\infty} \mathbb{P}[Y > k]$$

Tail Formula for Expectation.

In the geometric case:

$$\begin{aligned} \mathbb{E}[Y] &= \sum_{k=0}^{\infty} \mathbb{P}[Y > k] = \mathbb{P}[Y > 0] + \mathbb{P}[Y > 1] + \dots \\ &= q + q^2 + \dots + q^k + \dots \\ &= q(1 + q + \dots + q^{k-1} + \dots) \\ &= q \cdot \frac{1}{1-q} = \frac{q}{p} \end{aligned}$$



Variance.

Def'n. The **variance** of a random variable Y is defined as

$$\text{Var}[Y] := \mathbb{E}[(Y - \mathbb{E}[Y])^2] \quad \text{if "finite"}$$

The **standard deviation** of Y is defined as

$$\text{SD}[Y] = \sqrt{\text{Var}[Y]}$$

Formula.

$$\text{Var}[Y] = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2$$

$$\rightarrow: \mu_Y := \mathbb{E}[Y]$$

$$\text{Var}[Y] = \mathbb{E}[(Y - \mu_Y)^2]$$

$$= \mathbb{E}[Y^2 - 2Y\mu_Y + \mu_Y^2]$$

linearity of $\mathbb{E}$

$$\begin{aligned}\text{Var}[Y] &= \mathbb{E}[Y^2] - 2\overbrace{\mu_Y \mathbb{E}[Y]}^{\mu_Y^2} + \mu_Y^2 \\ &= \mathbb{E}[Y^2] - 2\mu_Y^2 + \mu_Y^2 \\ &= \mathbb{E}[Y^2] - \mu_Y^2\end{aligned}$$



Example. Bernoulli. $Y \sim \mathcal{B}(p)$

$$\text{Var}[Y] = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2$$

$$\begin{array}{l} Y \sim \begin{cases} 1 & w/p \\ 0 & w/q \end{cases} \\ Y^2 \sim Y \end{array}$$

$$\rightarrow \mathbb{E}[Y] - (\mathbb{E}[Y])^2 = p - p^2 = p(1-p)$$

$$\text{Var}[Y] = pq$$